Weight Optimization of the Bridge Girder of an Overhead Crane

This article is about applying the capabilities of the optimization module of the Python Library Scipy to solve a typical mechanical problem, i.e. weight optimization of a bridge girder under a given load considering strength and geometric constraints. Additionally, it gives some advice as how to interpret the results in order to derive a final design.

The emphasis lies on the optimization procedure. Thus, this article does not provide the state of art of how to perform a stress analysis of a bridge girder of an overhead crane, since, e.g., it does not consider any standards. Furthermore, simplified assumptions are made, which are not validated.

1. Definition of the Problem

Figure 1.1 shows a principal sketch of the overhead crane. The bridge girder with span L should have a T-shape. The hoist with mass m_K travels directly on the flange of the section. The load to be lifted has mass m_L.

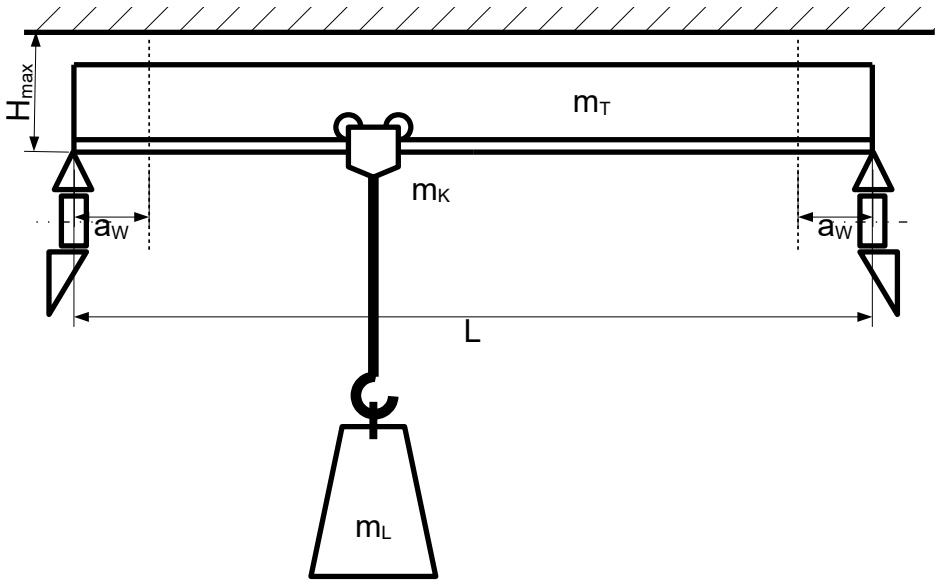


Figure 1.1: Sketch of the overhead crane

The crane operates in an engine house for repair and maintenance. Therefore, it is used seldom and thus the crane can be simply designed for static strength.

The task of the optimization problem is to find the dimensions of the section area of the bridge girder giving minimum weight of the crane under the following constraints:

- The maximum girder height must be smaller than H_{max} in order to avoid contact with the roof.
- The girder must have sufficient static strength.
- The maximum deflection of the girder has to be smaller or equal to L/500.
- The trolley requires a minimum width b_{Rmin} for operation (distance between web and outer edge of the flange, see figure 1.2).
- The thickness of the flange must be greater equal a minimum value t_{Gmin} to cover stresses induced by the wheels of the trolley. If distance between web and outer edge is grater than b_{Rmin} the permissible height of the flange must be increased by the ratio of the increase of the width and b_{Rmin} squared.
- The mass of trolley and hoist is given for minimum width b_{Rmin}. For larger distance b_R the mass has to be increased by $0.1 \cdot m_{K0} \cdot (b_G - b_{Gmin})$ considering that b_R is proportional to b_G. The minimum thickness of the flange applies for a flange width $b_G = b_{Gmin}$. If the flange width is larger than b_{Gmin} the required minimum thickness t_{Gmin} increases acc. to the ratio of actual and minimum width squared. This is chosen to roughly cover additional stresses induced by a wider hoist, such as bending stress of the flange and contact stress (due to higher mass of the hoist).
- Flange and web are going to be welded together. Welding procedure requires small differences regarding the thickness of both parts. Therefore, the thicknesses of web and flange must not differ by a factor grater than 1,6.

Distance a_W is the minimum distance the cable can have with respect to the wall.

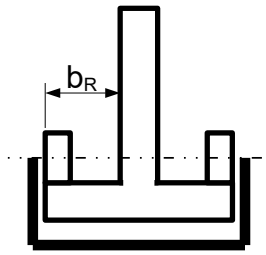


Figure 1.2: Distance b_R required by trolley

2. Designation of general Dimensions and Parameters

The principal dimensions are given as follows:

- L = 6m
- a_W = 0.4m
- H_{max} = 0.6m

The mass to be lifted is $m_L = 10t$. The mass of the hoist is $m_{K0} = 2t$ for $b_{Gmin} = 0.3m$ (considers required b_{Rmin}). The minimum permissible thickness of the flange is $t_{Gmin} = 16mm$.

The material is (weldable) construction steel S355 with the following properties:

- modulus of elasticity: E = 210000MPa
- density: $\rho = 7850kg/m^3$
- yield strength:
  - $R_{eh} = 355MPa$ (material thickness $\leq 16mm$)
  - $R_{eh2} = 345MPa$ (material thickness > 16mm, but $\leq 40mm$)

The following safety factors have to be applied:

- $j_{stens} = 1.5$ (tension stress, shear stress)
- $j_{scomp} = 1,1$ (compression stress)

Acceleration due to gravity is $g = 9,81m/s^2$.

A lifting operation may cause some dynamic load effect, which adds to weight load. This effect is simply covered by a load factor $f_L = 1.5$, which applies to the lifting mass only.

3. Formulation of the optimization problem

3.1 General Remark

A numerical problem may arise, if objective function and constraints have different order of magnitudes. A critical constraint having small order of magnitude may be interpreted as feasible when compared to a constraint giving high order of magnitude, though it may be violated instead. E.g., this can arise, if the constraints contain displacements (small order of magnitude, depending on unit) as well as stresses (high order of magnitude, depending on unit).

Therefore objective function and constraints are scaled by a typical value, such that the results becomes approx. one (i.e. order of magnitude equals 0). In addition, this approach has the advantage, that problems with compatibility of units are reduced or even avoided.

3.2 Design variables

Here only the dimensions of the T-section are allowed to be changed in order to achieve the best possible configuration.

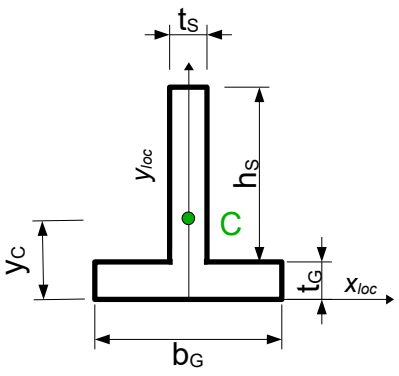


Figure 3.1: Dimensions of the T-Section

Acc. to figure 3.1 the design variables are:

- width of the flange b_G
- thickness of the flange t_G
- height of the web h_S
- thickness of the web t_S

Thus, the problem has four design variables.

3.3 Objective function

The goal is to find a configuration giving minimum weight, i.e. mass, of the overhead crane. The total mass is the sum of the masses of the bridge girder m_T and the hoist m_K here:

$$objective\ function = m_T + m_K$$

Both depend on the the design variables as follows

bridge girder: $m_T = \rho \cdot A_T \cdot L = \rho \cdot \left(h_S \cdot t_S + b_G \cdot t_G \right) \cdot L$

hoist: $m_K = m_{K0} \cdot \left(1 + 0.1 \cdot \frac{b_G}{b_{Gmin}} \right) \quad \forall\, b_G \geq b_{Gmin}$

The objective function is scaled (i.e. divided) by the lifting mass m_L (which remains constant during the optimization process).

3.4 Constraints

Scipy differs between bounds and constraints, though both give restrictions in order to meet additional requirements. Constraints are either equality constraints or inequality constraints. Here only inequality constraints occur.

The solver requires the constraints to be defined in the form $value\ of\ constraining\ function \geq 0$ if they constraint is fulfilled. I.e., if the value is lower than zero, the constraint will be violated. If the value is equal to zero, the constraint is active, however, the underlying requirement is just met. Since the constraints are to be scaled, here something like a rate of utilization is determined. A typical example is the ratio of actual and permissible stress. It takes values between 0 and 1, if requirements are met or is larger than 1 if requirements are not met. Therefore, 1 minus the rate of utilization defines the constraints as required by the solver:

$$1 - rate\ of\ utilization \geq 0$$

Note, if the basic constraint is given in the form “lower equal” it can be transformed to the form “greater equal” by multiplying the inequality by -1.

3.4.1 Bounds

Bounds define the value range of the design variables by giving a minimum and maximum value each design variable can take. E.g. bounds define a feasible range which geometrical dimensions can have, or at least ensure that the variable is going to be positive. Bounds can be handled as constraints as well. However, then normally two constraints per design variable have to be defined in order to have the same effect.

Furthermore, bounds can limit design variables such that functions used during the optimization process remain within a permissible range. Here, (for convenience) the yield strength is given for a thickness up to 40mm (s. section 2). Thus, all thicknesses used here have to be confined to 40mm in order to obtain a valid yield strength. However, if such a constraint leads to numerical problems, the range of the yield strength may be enlarged to cover thicknesses over 40mm. This may be recommendable if one or both thicknesses become 40mm, i.e. lie directly at the boundary.

The minimum thickness is set to 5mm (regarding the welding operation). However, the minimum thickness of the flange is t_{Gmin} since it serves as a rail.

Minimum width of the flange is b_{Gmin}, maximum width is set $3 \cdot b_{Gmin}$, what is somehow arbitrary, since it is expected that the solver tries to avoid large b_G, which adds to both mass of the girder and hoist. Thus it is not to be expected that b_G reaches this maximum value. However, the maximum value should be large enough here, such it does not unnecessarily confine the solution.

Minimum height of the web is set to 100mm, maximum height is set to H_{max} (ignoring any contribution from the flange here). The former is arbitrary as well, but sufficiently high, since the height of the web is expected to have a significant influence on strength and stiffness.

Finally, the bounds can be summarized as:

$$b_{Gmin} \le b_G \le 3 \cdot b_{Gmin}$$
$$16\,mm \le t_G \le 40\,mm$$
$$100\,mm \le h_S \le H_{max}$$
$$5\,mm \le t_S \le 40\,mm$$

Bounds can not be scaled.

3.4.2 Maximum Height of the Girder

The height of the girder is the sum of the web height and thickness of the flange. The sum is divided by H_{max} giving a value lower equal one, if the constraint will be met. Thus the constraint becomes:

$$1 - \frac{h_S + t_G}{H_{max}} \ge 0$$

3.4.3 Stresses at the Girder

The bridge girder is a beam simply supported at both ends. It can be considered as determinate allowing to derive forces and moments directly from equilibrium conditions or even more convenient from textbooks, like (2), if a similar reference case exists.

The constraints are formulated with bending stress and shear stress(see figure 3,2), though the later is less critical here, since the girder can be considered as slender and the material (steel) is less prone to shear stress.

Other influence on stress are not taken into account, like local stresses induced by the wheels of the hoist. Nonetheless, it is taken roughly into account confining the minimum thickness of the flange dependent on the weight of the hoist and the width of the flange.

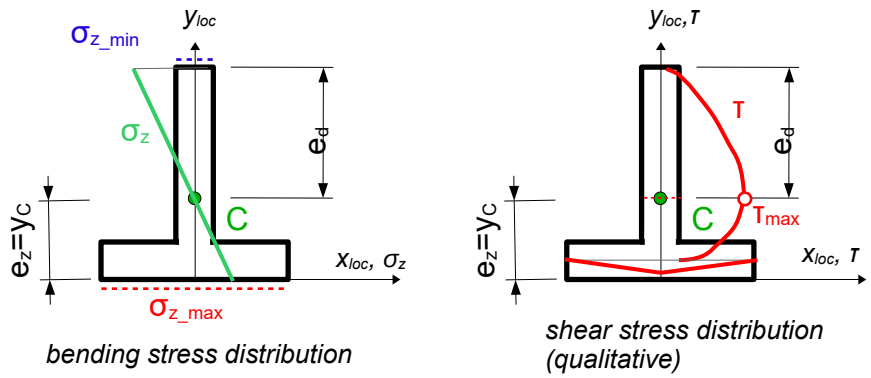


Figure 3.2: Bending stress σ_z and shear stress τ distribution

Maximum bending moment arises here if the hoist (with lifting mass) is at the mid position. However, the weight load of the girder has to be considered as well (see fig. 3.3). Thus, the maximum bending moment due to weight loads becomes:

$$M_{Bmax}=\left[\frac{1}{4}\cdot\left(f_L\cdot m_L+m_K\right)+\frac{1}{8}\cdot\rho\cdot A\cdot L\right]\cdot g\cdot L$$

A is the cross section of the girder, which depends on the actual configuration and thus on the actual values of the design variables.

Maximum transverse load occurs, when the hoist is near the wall, i.e. it has minimum distance a_W with respect to the wall (see figure 1.1). The maximum transverse load becomes

$$F_{max}=\left(f_L\cdot m_L+m_K\right)\cdot\frac{L-a_w}{L}+\frac{1}{2}\cdot\rho\cdot A\cdot L$$

For convenience, the transverse load due to girder weight is set equal to the bearing load, which is slightly higher than the precise value.

The section properties can be taken from a textbook as well (e.g. (2) or (3)), since a T-section is a standard section. Depending on the design variables one obtains the following properties:

total section area: $A = A_G + A_S = b_G \cdot t_G + h_S \cdot t_S$

A_G is the section area of the flange, A_S that of the web.

vertical location of the centroid (point C in figure 2.1): $y_C = \frac{b_G \cdot t_G^2 + t_S \cdot h_S \cdot (2 \cdot t_S + h_S)}{2 \cdot (t_G \cdot b_G + t_S \cdot h_S)}$

moment of inertia of flange (with respect to centroid of flange): $I_G = \frac{b_G \cdot t_G^3}{12}$

moment of inertia of web (with respect to centroid of web): $I_S = \frac{t_S \cdot h_S^3}{12}$

total moment of inertia: $I_{tot} = I_G + A_G \cdot \left(y_C - \frac{t_G}{2} \right)^2 + I_S + A_S \cdot \left(t_G + \frac{h_S}{2} - y_C \right)^2$

The distance from the centroid to the outmost “fiber” of the flange equals y_C, the distance from the centroid to the outmost “fiber” of the web is $ed = t_G + h_S - y_C$.

Finally, bending stresses can be generally determined as follows:

$$\sigma_B = \frac{M_B}{W} = \frac{M_B}{I_{tot}} \cdot y$$

y is the distance between centroid and “fiber” under consideration.

The maximum tensile stress due to bending occurs at the bottom of the girder here, i.e. at the flange. The constraint to confine maximum tensile stress becomes:

$$1 - \frac{j_{stens} \cdot M_{Bmax} \cdot y_C}{I_{tot} \cdot R_{eG}} \geq 0$$

$$R_{eG} = R_{eh} \quad if\ t_G \leq 16\,mm,\ R_{eG} = R_{eh2} \quad if\ t_G > 16\,mm \wedge t_G \leq 40\,mm$$

In case of compressive stress, which occurs at the web, the constraint becomes:

$$1 - \frac{j_{scomp} \cdot M_{Bmax} \cdot e_d}{I_{tot} \cdot R_{eS}} \geq 0$$

$$R_{eS} = R_{eh} \quad if\ t_S \leq 16\,mm,\ R_{eS} = R_{eh2} \quad if\ t_S > 16\,mm \wedge t_S \leq 40\,mm$$

Note that this constraint is set up considering solely the amount of the compressive stress (without considering the negative signs in detail).

Maximum shear stress is quite small, where maximum bending stresses occur. Therefore, it is not taken into account at these locations . Otherwise an equivalent stress had to be determined. The maximum shear stress occurs at point C, where bending stress is zero. However, shear stress can be considered as uncritical here. Thus, for convenience it is assumed that shear stress is constant and it is determined solely with the area of the web. Thus the constraint confining maximum shear stress becomes:

$$1-\frac{j_{stens}\cdot F_{max}\cdot\sqrt{3}}{A_S\cdot R_{eS}}\geq 0$$

$$R_{eS}=R_{eh}\ \ if\ t_S\leqslant 16\,mm\,,\ R_{eS}=R_{eh2}\ \ if\ t_S>16\,mm\wedge t_S\leq 40\,mm$$

The factor $\sqrt{3}$ follows from the equivalent von Mises stress.

Compressive stress and shear stress may give rise to buckling, which is not taken into account here.

3.4.4 Maximum Deflection

Maximum deflection occurs when the hoist (with lifting mass) is at the mid position, as with maximum bending moment. Since the girder can be considered as a slender beam, deflection due to shear force is neglected.



Figure 3.3: Bending deflection - influence of point and distributed load

Here maximum deflection is caused by the weight forces of hoist, lifting mass and girder (see maximum bending moment in section 3.4.3). The later is due to a distributed load whereas the former arise due to point loads. The deflections of each load condition can be superposed to the resulting deflection, giving the following amount:

$$u_{max}=u\left(f_L\cdot m_L+m_K\right)+u\left(m_T\right)=\left[\frac{f_L\cdot m_L+m_K}{48}+\frac{5\cdot\rho\cdot A\cdot L}{384}\right]\cdot\frac{g\cdot L^3}{E\cdot I_{tot}}$$

The load factor is considered as well (i.e. $F'_L = F_L \cdot f_L$), since it gives rise to additional deflection (e.g. static deflection plus vibration amplitude). Finally, the constraint inequality becomes:

$$1-\frac{500\cdot u_{max}}{L}\geq 0$$

3.4.5 Minimum Thickness of the Flange

As given in section 1 the minimum thickness of the flange depends on the width of the flange. The following is recommended:

$$t_{Gmin}^{act}=\left[\frac{b_G}{b_{Gmin}}\right]^2\cdot t_{Gmin}$$

The constraint follows as

$$1-\frac{t_{Gmin}^{act}}{t_G}=1-\left[\frac{b_G}{b_{Gmin}}\right]^2\cdot\frac{t_{Gmin}}{t_G}\geq 0$$

3.4.6 Thickness Ratio of Flange and Web

The ratio of the maximum and the minimum of both thicknesses (flange and web) must not exceed 1.6. However, a priori it is not defined if the thickness of the flange or the thickness of the web is the greatest. Thus the constraint can be either defined using a logical query or two constraint can be defined with different denominators. Here the later is used:

$$1-\frac{t_G}{t_S \cdot 1.6} \geq 0$$

$$1-\frac{t_S}{t_G \cdot 1.6} \geq 0$$

4. Python Script

The python procedure consists of two modules in order to separate mechanical analysis and optimization. The first module (OverheadCrane.py) is used to define the functions for the analysis of the bridge girder. It is written such that it can be used as a script as well. Thus, a given configuration can be analyzed. Furthermore, it contains some plot routines in order to visualize the configuration.

The main script (OptimizationOhCrane.py) performs the optimization, the module providing the crane analysis is included. It sets up the optimization problem, solves it and prints the results.

Here the method 'trust-constr' (trust-region algorithm for constrained optimization) has performed the best. Acc. to (1) "it is the most versatile constrained minimization algorithm implemented in SciPy and the most appropriate for large-scale problems". The method COBYLA performed sufficiently good as well. SLSQP (Sequential Least SQuares Programming) converges to solutions, but often to somehow local optima. It may converge to the solution of 'trust-constr' as well, if the initial guess is sufficiently close to the optimum solution. The method COBYQA totally failed here.

The option "disp" is set to "true" providing additional information of the optimization process.

Both files can be found in (4).

5. Result of the Optimization

5.1 Initial Configuration

The initial configuration is set up with the following dimensions:

$b_G = 350\,mm$
$t_G = 22\,mm$
$h_S = 200\,mm$
$t_S = 16\,mm$

These values are chosen quite arbitrarily, such that some of the constraints are met, but not necessarily all.

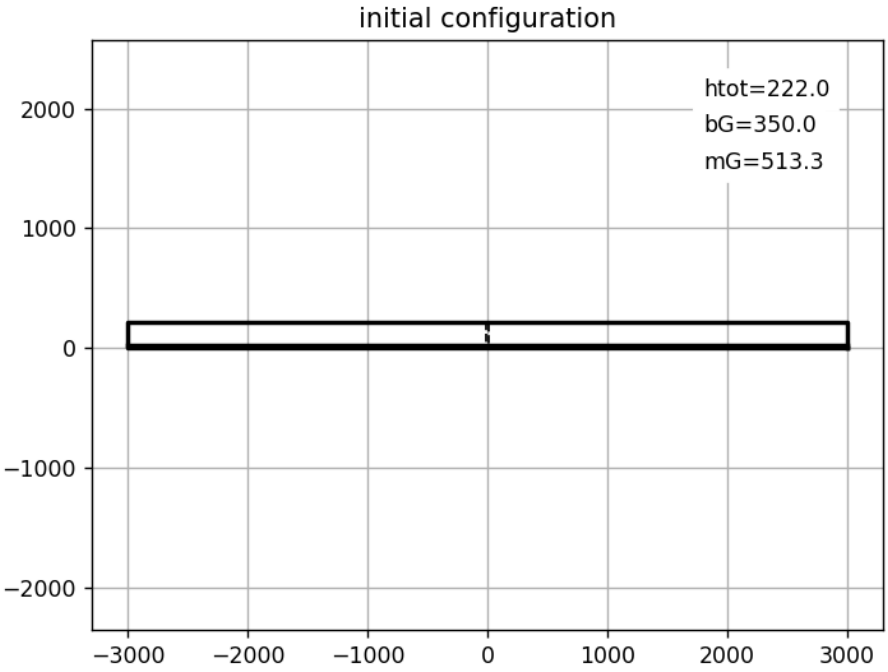


Figure 5.1 Plot of the initial Configuration of the Overhead Crane

Fig. 5.1 provides a plot of the initial configuration, fig 5.2 depicts the initial cross section.

h_{tot} is the total height, b_G is the width of the flange (design variable) and m_G is the mass of the girder.

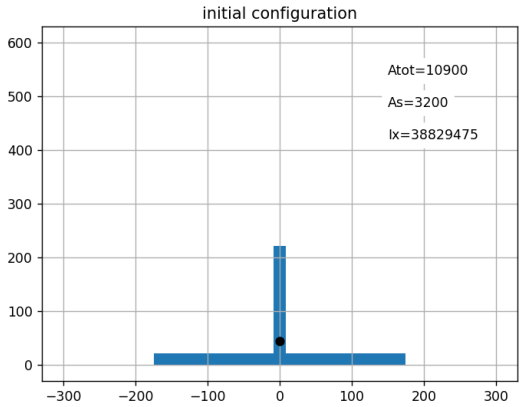


Figure 5.2: Plot of the Cross Section of the Initial Configuration

A_{tot} is the total section area, A_S is the section area of the web and I_x is the moment of inertia. The dot shows the location of the centroid of the cross section.

This initial configuration violates some of the constraints (stress constraints, deflection constraint), giving ratios of utilization clearly above one (see fig. 5.3). Therefore, this configuration does not give a useful design.

```
actual results for initial configuration:
weight force = 264870.0
max. force = 249730.17795
max.bending moment = 401082266.925
max. bending deflection =
147.90920634952434
utilizations:
tensile stress:  1.6312532378430113
compressive stress:  5.710345486393503
shear stress:  0.4759521967348338
max. deflection:  12.325767195793695
min. thickness flange:  0.9898989898989901
thickness ratio 1:  0.859375
thickness ratio 2:  0.45454545454545453
max. height:  0.37
```

Figure 5.3 Some Analysis Results of the Initial Configuration

5.2 Optimal Configuration

Figure 5.4 depicts the message of the optimizer.

```
gtol` termination condition is satisfied.
Number of iterations: 76, function evaluations: 455, CG iterations: 106, optimality: 9.48e-09, constraint
violation: 0.00e+00, execution time:  0.2 s.

         message: `gtol` termination condition is satisfied.
         success: True
          status: 1
             fun: 0.2475058788641939
               x: [ 3.000e+02  1.600e+01  5.284e+02  1.000e+01]
             nit: 76
            nfev: 455
            njev: 91
            nhev: 0
       cg_niter: 106
    cg_stop_cond: 4
            grad: [ 2.008e-02  1.413e-03  4.711e-05  2.489e-03]
  lagrangian_grad: [ 9.954e-12  7.239e-11 -9.482e-09  2.464e-10]
          constr: [array([ 5.527e-01]), array([ 8.449e-05]), array([ 8.177e-01]), array([ 2.108e-02]),
array([ 7.912e-05]), array([ 1.099e-04]), array([ 6.093e-01]), array([ 9.258e-02]), array([ 3.000e+02,
1.600e+01,  5.284e+02,  1.000e+01])]
```

Figure 5.4: Message of the Optimizing Module

First, it has to be checked, whether the optimizer has converged to a solution, which is the case here: success is true, ‘status’ and ‘cg_stop_cond’ (solver specific output) indicate, that a solution is found within the tolerance.

Furthermore, the output provides numerical values of the objective function (note that it is scaled by the lifting mass) and the design variables giving the optimum configuration.

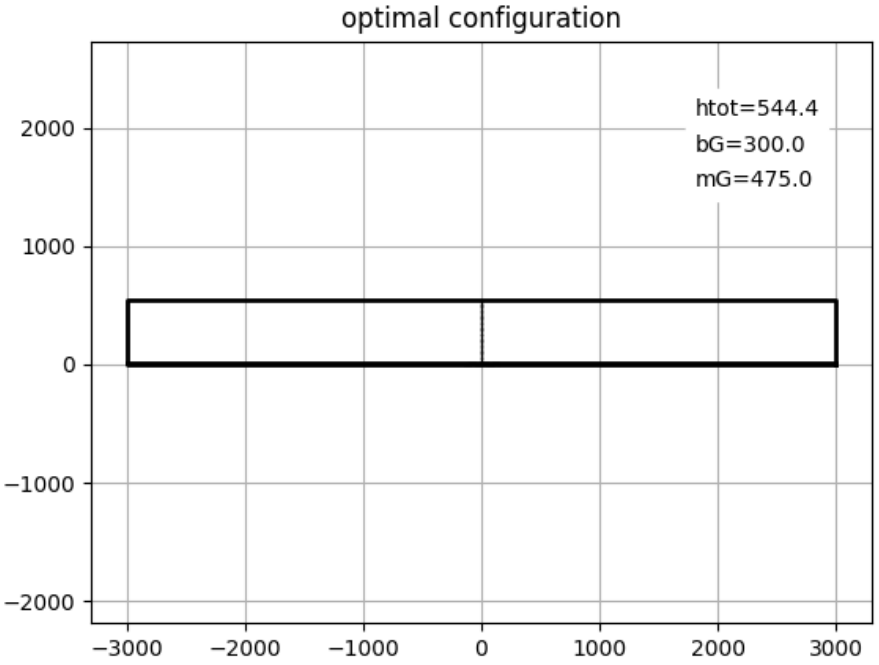


Figure 5.5 Plot of the Optimal Configuration of the Overhead Crane

Figure 5.5 shows the optimal configuration of the crane. The height of the girder is clearly higher as that of the initial configuration (see fig. 5.1), what is obviously a necessary approach to meet the requirements. The minimum useful mass of the girder is m_G = 475kg acc. to the result.

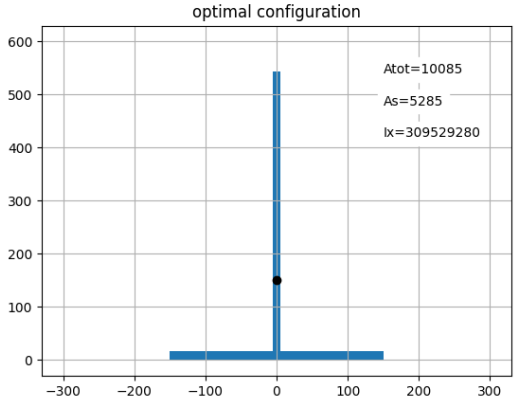


Figure 5.6: Plot of the Cross Section of the Optimal Configuration

Figure 5.6 depicts the optimal cross section. The thickness of the flange is greater than that of the web. The former contributes the most to the moment of inertia by its distance to the centroid squared (acc. to the parallel axis theorem).

The analysis results show, that the constraints are met by the optimal configuration: The utilizations practically do not exceed one.

```
actual results for optimal configuration:
weight force = 166770.13719992645
max. force = 157982.22281159018
max.bending moment = 253650347.9373779
max. bending deflection =
11.747060073955668
utilizations:
tensile stress:  0.4473244631642624
compressive stress:  0.9999155096411574
shear stress:  0.1822908805555093
max. deflection:  0.978921672829639
min. thickness flange:  0.9999208809941027
thickness ratio 1:  0.9998901478246842
thickness ratio 2:  0.39066791572036785
max. height:  0.9074175454394411
```

Figure 5.7 Some Analysis Results of the Optimal Configuration

Here the following constraints are active, i.e. these constraints are binding (utilizations are equal or nearly equal to one):

- permissible compressive stress
- minimum thickness of the flange
- thickness ratio

The amount of deflection is near its permissible value, but below the threshold, where the constraint becomes active.

Finally, the optimal section has the following dimensions:

$b_{Gopt} = 300\,mm$
$t_{Gopt} = 16.0\,mm$
$h_{Sopt} = 528.4\,mm$
$t_{Sopt} = 10{,}0\,mm$

The width b_G is equal to its lower bound (see section 3.4.1). This is due to the fact that the width has an influence on the mass of the hoist as well. Both thicknesses are equal to their lower bounds as well. The height of the web lies within the given bounds.

6. Some Remarks on Practical Implementation of the Optimum Configuration

The optimization delivers non-integer numbers as solution. This may not be convenient considering additional aspects, like costs, which often have a critical influence in general mechanical

engineering (other than in space technology). E.g., if semi-finished products are welded together to form the final product, they should have standard dimensions as far as possible. Otherwise, additional costs arise in production.

Here, the thicknesses should be integer values acc. to what can be delivered. Since they lie at their lower bounds, this is the case here. However, sheet metal may not available in consecutive thicknesses. Then the nearest available thickness to the optimum thickness has to be chosen. Care has to be taken, that the constraints remain unviolated.

The other variables need not be integer values here, but may simplify measurement during production, if they will be. E.g., h_S may be increased to 530mm with a negative tolerance. Its impact on weight should be small and it still lies within the height constraint.

The web of the cross section can be considered as a thin wide plate, what may give rise to buckling. Therefore, it may have to be checked in addition if the dimensions of the optimum configuration may cause instability. If it is critical or is expected to be critical, additional constraints have to be added to the definition of the optimization problem right from the beginning. However, stability effects are not considered here.

Theoretically, it may be possible to formulate such a problem as mixed integer programming in order to get required dimensions directly. If a semi-finished product is available in discrete, but not consecutive dimensions, the available dimensions may be put in an array. Then, the design variable is the index of the array and thus becomes a consecutive number. However, mixed integer programming is even more challenging.

7. Summary

Using the bridge girder of an overhead crane as an example this article shows how to formulate an optimization problem, which can be solved by the Python Package Scipy. The solution of the optimization problem requires to chose a solver capable to handle the constraints. Therefore, the output of the solver has to be thoroughly checked in order to ensure that it has found a useful configuration, which does not differ much from the global optimum.

Literature:

(1) Scipy User Guide – Optimization. 2025 Link: Optimization (scipy.optimize) — SciPy v1.16.2 Manual

(2) Schaeffler Technical Pocket Guide. 2025-03-14 Link: Schaeffler Technical Pocket Guide | Media Library | Schaeffler Germany

(3) Young, Budynas, Sadegh: Roark's Formulas for Stress and Strain. McGraw-Hill 2012

(4) Python Files Link: https://github.com/kroneaba/try_sympy_mechanics.git

(5) Grobblech 1.0038 aus Walzstahl | Thyssenkrupp Schulte. 2025